

-- ===== SAYLANI DATA ENGINEERING =====

-- HOMEWORK: CLASS 4 - MODIFYING DATA, DDL, DATA TYPES & CONSTRAINTS

-- Database: BikeStores Sample Database

-- Instructions: Write SQL statements to solve each problem below.

-- Answers will be provided in a separate file.

-- =====

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-- =====

-- SECTION A: DATA TYPES & CONSTRAINTS (Conceptual Questions)

-- =====

-- Q1: What data type would you use for a product's weight (e.g., 2.5 kg)?

weight decimal(5,3) 12345.999

ANS : DECIMAL(5,3)

-- Q2: In the sales.stores table, the zip_code is VARCHAR(5). Why not use INT?

ANS : Because ZIP/postal codes are:

identifiers, not numbers for calculation

We never:

- add ZIP codes
- subtract ZIP codes
- multiply ZIP codes

So don't need numeric datatype

-- Q3: Look at sales.orders.order_status. The comment says
1=Pending,2=Processing,3=Rejected,4=Completed.

-- Is TINYINT a good choice? Why not use INT?

select * from [sales].[orders]

--yes, tinyint is a good choice

ANS : TINYINT is a very good choice for order_status. Because order statuses only contain small fixed values Like as Pending = 1, Processing = 2 , Rejected = 3, Completed = 4 .

-- Q4: If you add a CHECK constraint that rating must be BETWEEN 1 AND 5, what happens if you try to INSERT rating = 0?

ANS : SQL Server will reject the INSERT.
0 is not between 1 and 5

The CHECK constraint validates data before insertion and enforces business rules / valid data

If we try to insert rating = 0, the INSERT operation will fail because it violates the CHECK constraint BETWEEN 1 AND 5.

-- Q5: Why does sales.staffs have UNIQUE constraint on email but not on phone?

ANS : Database Design Rule :

UNIQUE = stable + reliable + truly identifying

Email fits → unique identity

Phone does NOT → can change / shared

Email has a UNIQUE constraint because it is a stable and reliable identifier for each staff member, commonly used for login and communication. Phone numbers are not unique because they can change, be shared among multiple people, or be entered in different formats, making them unreliable for enforcing uniqueness.

-- =====

-- SECTION B: DDL (CREATE, ALTER, DROP)

-- =====

```
-- Q6: Create a new table called sales.loyalty_programs with the following columns:
--   - program_id (INT, auto-increment starting 1, PRIMARY KEY)
--   - program_name (VARCHAR(100), NOT NULL, UNIQUE)
--   - discount_rate (DECIMAL(3,2), NOT NULL, DEFAULT 0.05, CHECK between 0.00 and 0.50)
--   - start_date (DATE, NOT NULL, DEFAULT GETDATE())
--   - end_date (DATE, NULL)
```

```
CREATE TABLE sales.loyalty_programs (
    program_id INT IDENTITY(1,1) NOT NULL,
    program_name VARCHAR(100) NOT NULL UNIQUE,
    discount_rate DECIMAL(3,2) NOT NULL
        DEFAULT 0.05
        CHECK (discount_rate BETWEEN 0.00 AND 0.50),
    start_date DATE NOT NULL DEFAULT GETDATE(),
    end_date DATE NOT NULL DEFAULT GETDATE()
);
```

SQLQuery2.sql...fujitsu (66))*

```
1  
2  
3 CREATE TABLE sales.loyalty_programs (  
4     program_id INT IDENTITY(1,1) PRIMARY KEY,  
5     program_name VARCHAR(100) NOT NULL UNIQUE,  
6     discount_rate DECIMAL(3,2) NOT NULL DEFAULT 0.05  
7         CHECK (discount_rate BETWEEN 0.00 AND 0.50),  
8     start_date DATE NOT NULL DEFAULT GETDATE(),  
9     end_date DATE NULL  
10 );  
11  
12 SELECT * FROM sales.loyalty_programs;
```

100 %

✓ No issues found



Results



Messages

program_id	program_name	discount_rate	start_date	end_date
------------	--------------	---------------	------------	----------

✓ Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM)

- sales.customers
- sales.loyalty_programs
 - Columns
 - program_id (PK, int, not null)
 - program_name (varchar(100), not null)
 - discount_rate (decimal(3,2), not null)
 - start_date (date, not null)
 - end_date (date, null)
 - Keys
 - Constraints
 - Triggers

-- Q7: Add a new column 'loyalty_program_id' (INT, NULL) to the sales.customers table.

alter table sales.customers

add [loyalty_program_id] INT null;

The screenshot shows a SQL IDE window with a query editor at the top and a results pane at the bottom. The query editor contains the following SQL code:

```
1  
2  
3 ALTER TABLE sales.customers  
4 ADD loyalty_program_id INT NULL;  
5  
6 SELECT * FROM sales.customers
```

The results pane shows a table with 11 rows and 11 columns. The columns are: customer_id, first_name, last_name, phone, email, street, city, state, zip_code, and loyalty_program_id. The data is as follows:

	customer_id	first_name	last_name	phone	email	street	city	state	zip_code	loyalty_program_id
1	1	Debra	Burks	NULL	debra.burks@yahoo.com	9273 Thome Ave.	Orchard Park	NY	14127	NULL
2	2	Kasha	Todd	NULL	kasha.todd@yahoo.com	910 Vine Street	Campbell	CA	95008	NULL
3	3	Tameka	Fisher	NULL	tameka.fisher@aol.com	769C Honey Creek St.	Redondo Beach	CA	90278	NULL
4	4	Daryl	Spence	NULL	daryl.spence@aol.com	988 Pearl Lane	Uniondale	NY	11553	NULL
5	5	Charolette	Rice	(916) 381-6003	charolette.rice@msn.com	107 River Dr.	Sacramento	CA	95820	NULL
6	6	Lyndsey	Bean	NULL	lyndsey.bean@hotmail.com	769 West Road	Fairport	NY	14450	NULL
7	7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215	NULL
8	8	Jacqueline	Duncan	NULL	jacqueline.duncan@yahoo.com	15 Brown St.	Jackson Heights	NY	11372	NULL
9	9	Genoveva	Baldwin	NULL	genoveva.baldwin@msn.com	8550 Spruce Drive	Port Washington	NY	11050	NULL
10	10	Pamela	Newman	NULL	pamela.newman@gmail.com	476 Chestnut Ave.	Monroe	NY	10950	NULL
11	11	Deshawn	Mendoza	NULL	deshawn.mendoza@yahoo.com	8790 Cobblestone Street	Monsey	NY	10952	NULL

The status bar at the bottom indicates "Query executed successfully." and "DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fujits... | BikeStores | 00:00:00 | Row: 1, Col: 1 | 1,445 rows".

The left pane shows the database structure:

- production.stocks
- sales.customers
 - Columns
 - customer_id (PK, int, not null)
 - first_name (varchar(255), not null)
 - last_name (varchar(255), not null)
 - phone (varchar(25), null)
 - email (varchar(255), not null)
 - street (varchar(255), null)
 - city (varchar(50), null)
 - state (varchar(25), null)
 - zip_code (varchar(10), null)
 - loyalty_program_id (int, null)
 - Keys
 - Constraints

-- Q8: Add a FOREIGN KEY constraint to sales.customers.loyalty_program_id that references
-- sales.loyalty_programs.program_id.

The screenshot displays a SQL IDE window titled "SQLQuery2.sql...fujitsu (66))". The main editor shows a SQL query with line numbers 1 through 6. The query is:
1
2
3 ALTER TABLE sales.customers
4 ADD CONSTRAINT FK_customers_loyalty_program
5 FOREIGN KEY (loyalty_program_id)
6 REFERENCES sales.loyalty_programs(program_id);
Below the editor, a status bar indicates "100 %", a green checkmark, and "No issues found". A "Messages" pane shows the message "Commands completed successfully." and the completion time "2026-05-25T11:27:15.9327445+05:00". At the bottom, a database schema tree is visible, showing the "sales" database with tables "sales.customers" and "sales.loyalty_programs". The "sales.loyalty_programs" table is expanded, showing its columns: "program_id (PK, int, not null)", "program_name (varchar(100), not null)", "discount_rate (decimal(3,2), not null)", "start_date (date, not null)", and "end_date (date, null)". There are also sections for "Keys" and "Constraints" which are currently collapsed.

```
1  
2  
3 ALTER TABLE sales.customers  
4 ADD CONSTRAINT FK_customers_loyalty_program  
5 FOREIGN KEY (loyalty_program_id)  
6 REFERENCES sales.loyalty_programs(program_id);
```

100 % No issues found

Messages
Commands completed successfully.
Completion time: 2026-05-25T11:27:15.9327445+05:00

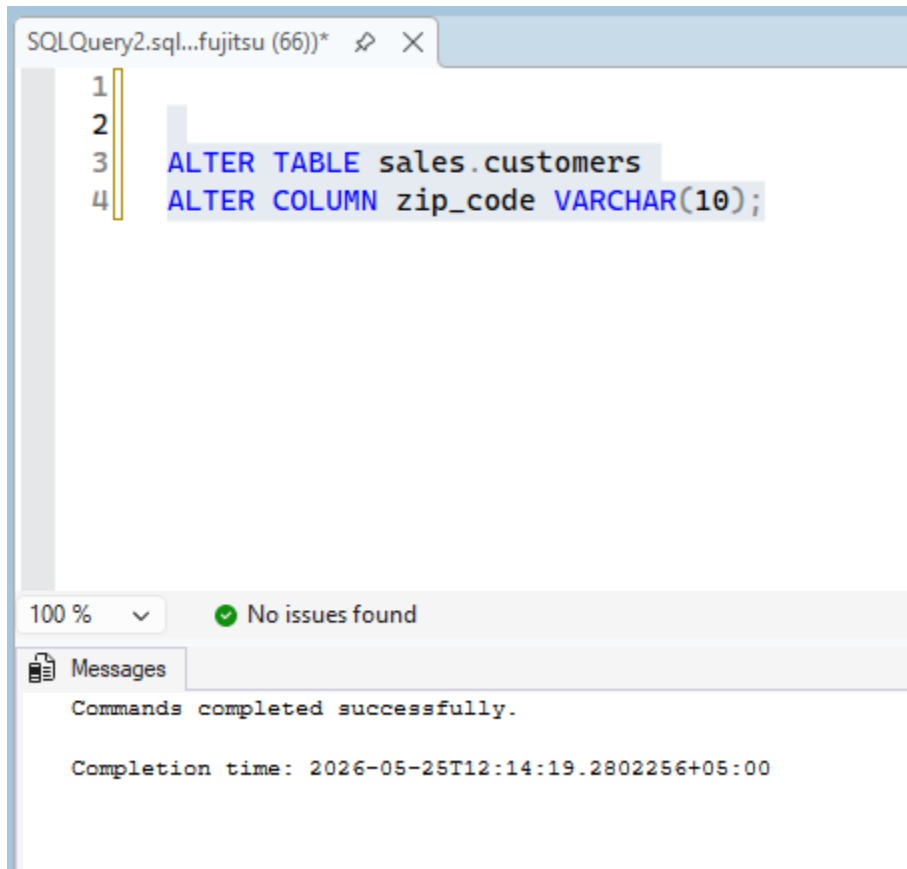
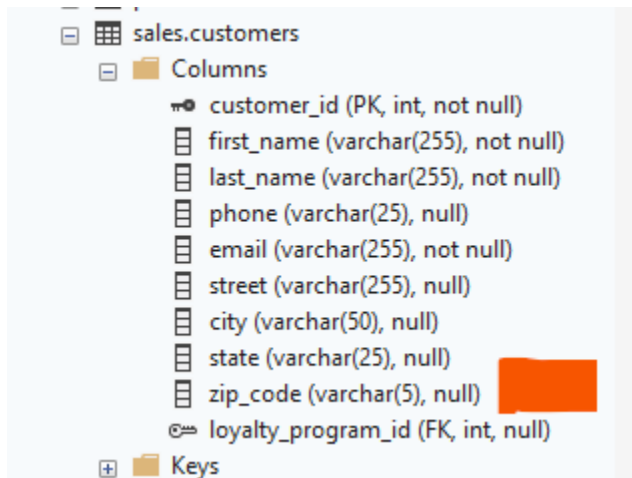
sales.customers
sales.loyalty_programs
Columns
program_id (PK, int, not null)
program_name (varchar(100), not null)
discount_rate (decimal(3,2), not null)
start_date (date, not null)
end_date (date, null)
Keys
Constraints

-- Q9: Change the data type of sales.customers.zip_code from VARCHAR(5) to VARCHAR(10).

```
ALTER TABLE sales.customers
```

ALTER COLUMN zip_code VARCHAR(10);

BEFORE :



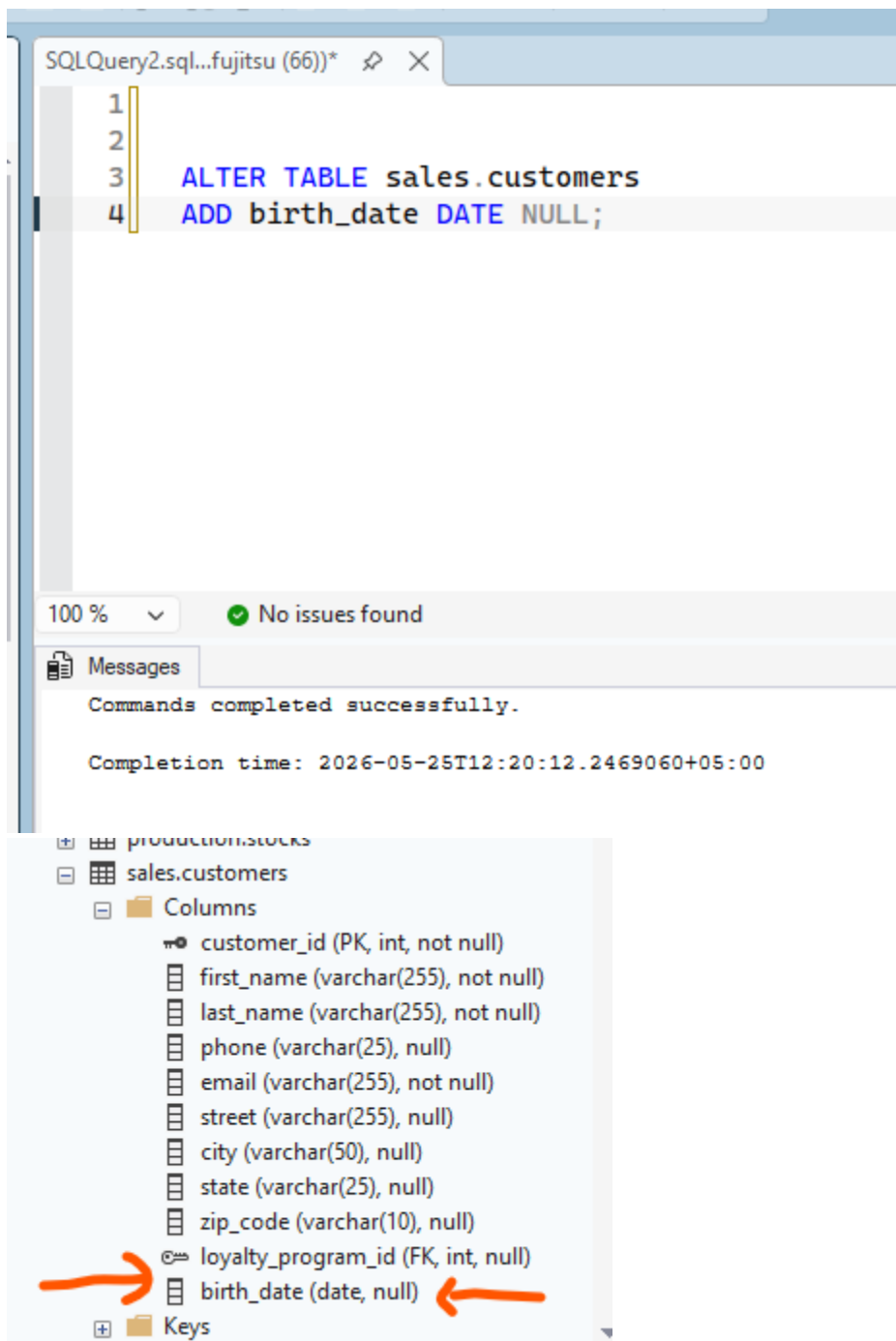
AFTER :

sales.customers
Columns
customer_id (PK, int, not null)
first_name (varchar(255), not null)
last_name (varchar(255), not null)
phone (varchar(25), null)
email (varchar(255), not null)
street (varchar(255), null)
city (varchar(50), null)
state (varchar(25), null)
zip_code (varchar(10), null)
loyalty_program_id (FK, int, null)
Keys

-- Q10: Drop the column 'birth_date' from sales.customers (first add it if it doesn't exist, then drop it).

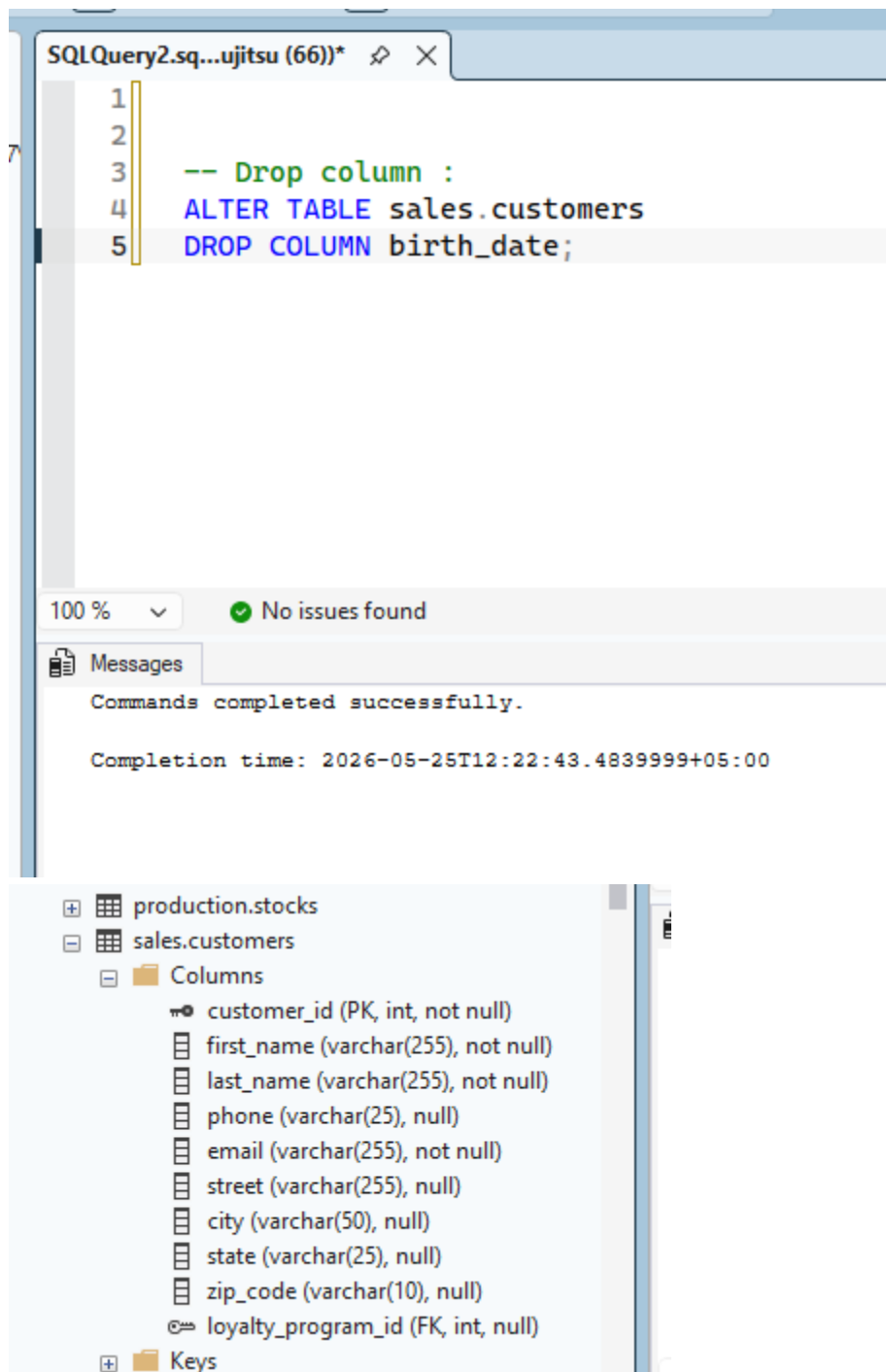
```
alter table sales.customers
```

```
add birth_date date null;
```



alter table sales.customers

drop column birth_date



- Q11: Create a new table production.product_reviews with appropriate columns and constraints:
- - review_id (PK, auto-increment)
 - - product_id (FK to production.products)

```
-- - customer_id (FK to sales.customers)
-- - rating (TINYINT, 1-5)
-- - review_text (VARCHAR(1000))
-- - review_date (DATE, default today)
```

```
SQLQuery2.sql...ujitsu (66))* X
2 CREATE TABLE production.product_reviews (
3
4     review_id INT IDENTITY(1,1) PRIMARY KEY,
5
6     product_id INT NOT NULL,
7     customer_id INT NOT NULL,
8
9     rating TINYINT NOT NULL
10     CHECK (rating BETWEEN 1 AND 5),
11
12     review_text VARCHAR(1000),
13
14     review_date DATE DEFAULT GETDATE(),
15
16     CONSTRAINT FK_product_reviews_products
17     FOREIGN KEY (product_id)
18     REFERENCES production.products(product_id),
19
20     CONSTRAINT FK_product_reviews_customers
21     FOREIGN KEY (customer_id)
22     REFERENCES sales.customers(customer_id)
23 )
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-05-25T12:28:32.6665678+05:00

100 % No issues found

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESK

- Graph Tables
- production.brands
- production.categories
- production.product_reviews
 - Columns
 - review_id (PK, int, not null)
 - product_id (FK, int, not null)
 - customer_id (FK, int, not null)
 - rating (tinyint, not null)
 - review_text (varchar(1000), null)
 - review_date (date, null)
 - Keys
 - Constraints

-- =====

-- SECTION C: INSERT STATEMENTS

-- =====

-- Q12: Insert a new brand called 'Santa Cruz' into production.brands.

SET IDENTITY_INSERT production.brands on;

INSERT INTO production.brands

(BRAND_ID, BRAND_NAME)

VALUES (9, 'Trek')

The screenshot shows a SQL Query Editor window titled "SQLQuery2.sql...ujitsu (66)". The query is as follows:

```
1
2  -- if brand_id auto-increment (IDENTITY) hai, then normally we don't manually insert ID.
3  --IDENTITY_INSERT ON ke baad usually OFF karte hain.
4  --system force to manual ID insertion:
5  -- simple query is :
6
7  SET IDENTITY_INSERT production.brands ON;
8
9  INSERT INTO production.brands (brand_id, brand_name)
10 VALUES (9, 'Santa Cruz');
11
12 SET IDENTITY_INSERT production.brands OFF;
13
14 SELECT * FROM production.brands;
15
16
17
```

The status bar at the bottom indicates "100 %", "No issues found", and "Ln: 10, Ch: 26 | T".

Below the query editor, the "Results" tab is active, displaying a table with the following data:

	brand_id	brand_name
2	2	Haro
3	3	Heller
4	4	Pure Cycles
5	5	Ritchey
6	6	Strider
7	7	Sun Bicycles
8	8	Surly
9	9	Santa Cruz

The status bar at the bottom of the Results tab indicates "Query executed successfully." and "DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fujitsu... | BikeStores | 00:00:00 | Row:".

```
SELECT * FROM production.brands;
```

```
=====
```

```
-- Q13: Insert three new categories at once: 'Mountain', 'Road', 'Hybrid'.
```

```
SELECT * FROM production.categories;
```

```
INSERT INTO production.categories
```

```
(category_name)
```

```
VALUES ('Mountain'), ('Road'), ('Hybrid')
```

SQLQuery2.sq...ujitsu (66))*

1

2

3

4

5

6

7

8

INSERT INTO production.categories (category_name)

VALUES

('Mountain'),

('Road'),

('Hybrid');

SELECT * FROM production.categories;

100 %

No issues found

Results

Messages

	category_id	category_name
2	2	Comfort Bicycles
3	3	Cruisers Bicycles
4	4	Cyclocross Bicycles
5	5	Electric Bikes
6	6	Mountain Bikes
7	7	Road Bikes
8	8	Mountain
9	9	Road

Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7A

-- Q14: Insert a new product with the following details:

-- product_name = 'Santa Cruz Bronson'

-- brand_id = (the brand_id of 'Santa Cruz' from Q12)

-- category_id = (category_id of 'Mountain' from Q13)

-- model_year = 2025

SQLQuery2.sql...ujitsu (66))*

```
1  
2  
3  
4  
5  
6 SELECT brand_id  
7 FROM production.brands  
8 WHERE brand_name = 'Santa Cruz';
```

100 %

✓ No issues found

Results

Messages

	brand_id
1	9

SQLQuery2.sql...ujitsu (66))*

```
1
2
3
4
5
6 SELECT category_id
7 FROM production.categories
8 WHERE category_name = 'Mountain';
```

100 % No issues found

Results Messages

	category_id
1	8

-- list_price = 4299.99

SQLQuery2.sql...ujitsu (66))*

```
1
2 INSERT INTO production.products
3 (product_name, brand_id, category_id, model_year, list_price)
4
5 VALUES (
6     'Santa Cruz Bronson',
7     (SELECT brand_id FROM production.brands WHERE brand_name = 'Santa Cruz'),
8     (SELECT category_id FROM production.categories WHERE category_name = 'Mountain'),
9     2025,
10    4299.99
11 );
12
13 select * FROM production.products
14 where product_name = 'Santa Cruz Bronson'
15
16 ;
```

100 % No issues found Ln: 10, Ch: 12 SPC

Results Messages

	product_id	product_name	brand_id	category_id	model_year	list_price
1	322	Santa Cruz Bronson	9	8	2025	4299.99

-- Q15: Copy all customers from California (state = 'CA') into a new table called sales.ca_customers_backup.

-- (Create the table first with the same structure as sales.customers)

SQLQuery1.sql...fujitsu (58))*

```
1
2 SELECT *
3 INTO sales.ca_customers_backup
4 FROM sales.customers
5 WHERE 1 = 0;
6
7
8 SELECT *
9 FROM sales.ca_customers_backup
10
```

100 % No issues found

Results Messages

customer_id	first_name	last_name	phone	email	street	city	state	zip_code	loyalty_program_id
-------------	------------	-----------	-------	-------	--------	------	-------	----------	--------------------

sales.ca_customers_backup

- Columns
 - customer_id (int, not null)
 - first_name (varchar(255), not null)
 - last_name (varchar(255), not null)
 - phone (varchar(25), null)
 - email (varchar(255), not null)
 - street (varchar(255), null)
 - city (varchar(50), null)
 - state (varchar(25), null)
 - zip_code (varchar(10), null)
 - loyalty_program_id (int, null)
- Keys

SQLQuery1.sql...ujitsu (58)*

```

6
7
8 SELECT *
9 FROM sales.ca_customers_backup;
10 -----
11
12 SET IDENTITY_INSERT sales.ca_customers_backup ON;
13
14 INSERT INTO sales.ca_customers_backup
15 (customer_id, first_name, last_name, state, city, email)
16 SELECT
17 customer_id, first_name, last_name, state, city, email
18 FROM sales.customers
19 WHERE state = 'CA';
20
21 SET IDENTITY_INSERT sales.ca_customers_backup OFF;

```

100 % No issues found Ln: 11, Ch: 1 TA

Results Messages

	customer_id	first_name	last_name	phone	email	street	city	state	zip_code	loyalty_program_id
1	2	Kasha	Todd	NULL	kasha.todd@yahoo.com	NULL	Campbell	CA	NULL	NULL
2	3	Tameka	Fisher	NULL	tameka.fisher@aol.com	NULL	Redondo Beach	CA	NULL	NULL
3	5	Charolette	Rice	NULL	charolette.rice@msn.com	NULL	Sacramento	CA	NULL	NULL
4	24	Corene	Wall	NULL	corene.wall@msn.com	NULL	Atwater	CA	NULL	NULL
5	30	Jamaal	Albert	NULL	jamaal.albert@gmail.com	NULL	Torrance	CA	NULL	NULL
6	31	Williemae	Holloway	NULL	williemae.holloway@msn.com	NULL	Oakland	CA	NULL	NULL
7	32	Araceli	Golden	NULL	araceli.golden@msn.com	NULL	Fullerton	CA	NULL	NULL

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7/fujitsu... BikeStores 00:00:00 Row: 1, C

-- =====

-- SECTION D: UPDATE STATEMENTS

-- =====

-- Q16: Update the phone number of customer with customer_id = 10 to '(555) 123-4567'.

SQLQuery1.sql...ujitsu (58))*

```
1
2
3 UPDATE sales.customers
4 SET phone = '(555) 123-4567'
5 WHERE customer_id = 10;
6
7
8 select * from sales.customers
9 WHERE customer_id = 10;
10
```

100 % No issues found Ln: 10, Ch: 1 TA

Results Messages

	customer_id	first_name	last_name	phone	email	street	city	state	zip_code	loyalty_program_id
1	10	Pamela	Newman	(555) 123-4567	pamela.newman@gmail.com	476 Chestnut Ave.	Monroe	NY	10950	NULL

-- Q17: Increase the list price of all products in the 'Road' category by 8%.

```
SQLQuery1.sql...ujitsu (58))* ✕
1
2
3 UPDATE p
4 SET p.list_price = p.list_price * 1.08
5 FROM production.products p
6 JOIN production.categories c
7     ON p.category_id = c.category_id
8 WHERE c.category_name = 'Road';
9 -----
10 select * from [production].[products]
11 where category_id = 9;
12 select * from [production].[categories];
13 -----
14
15 SELECT p.product_id,
16         p.product_name,
17         p.list_price,
18         c.category_name
19 FROM production.products p
20 JOIN production.categories c
21     ON p.category_id = c.category_id
22 WHERE c.category_name = 'Road';
```

Technical Logic issue :

"The query executed successfully, but returned zero rows because the production.products table contains no active records associated with category_id = 9 (Road)."

Results

Messages

product_id	product_name	brand_id	category_id	model_year	list_price
------------	--------------	----------	-------------	------------	------------

-- Q18: Mark all orders that have status = 4 (Completed) and shipped_date IS NULL

-- to set shipped_date = order_date + 3 days.

```
LQuery1.sql...ujitsu (58))* ✕ X
1
2
3 UPDATE sales.orders
4 SET shipped_date = DATEADD(DAY, 3, order_date)
5 WHERE order_status = 4
6     AND shipped_date IS NULL;
7
8
9 select shipped_date, order_status
10 from sales.orders
11 WHERE order_status = 4;
12
13
```

0 % No issues found

Results		Messages
shipped_date	order_status	
2016-01-03	4	
2016-01-03	4	
2016-01-03	4	
2016-01-05	4	
2016-01-06	4	
2016-01-05	4	

Query executed successfully. | DESKTOP-26P7AE7 (17.0 RTM) | DE

-- Q19: Set the manager_id of all staffs working at store_id = 1 to staff_id = 5

-- (assume staff_id 5 is the manager of that store).

SQLQuery1.sql...ujitsu (70))*

```
1
2
3
4 UPDATE sales.staffs
5 SET manager_id = 5
6 WHERE store_id = 1;
7
8
9 select manager_id ,store_id
10 from sales.staffs
11 WHERE store_id = 1;
```

100 % No issues found

Results Messages

	manager_id	store_id
1	5	1
2	5	1
3	5	1
4	5	1

Query executed successfully.

-- Q20: Update the discount for order_items where order_id = 100 and item_id = 2 to 0.15 (15%).



SQLQuery1.sql...ujitsu (70))*

```
1
2
3
4 UPDATE sales.order_items
5 SET discount = 0.15
6 WHERE order_id = 100
7     AND item_id = 2;
8
9
10 select order_id, item_id ,discount
11 from sales.order_items
12 WHERE order_id = 100
13     AND item_id = 2;
```

100 % No issues found

Results Messages

	order_id	item_id	discount
1	100	2	0.15

-- =====

-- SECTION E: DELETE STATEMENTS

-- =====

-- Q21: Delete the brand 'Santa Cruz' you inserted in Q12.

SQLQuery1.sql...ujitsu (70))*

```
1
2  --The Santa Cruz brand was successfully ir
3  --INSERT INTO production.brands (brand_nam
4  --VALUES ('Santa Cruz');
5
6
7  --Before deleting the brand, a safety chec
8  --whether the brand was being referenced i
9
10 SELECT brand_id,product_name
11 FROM production.products
12 --The query returned no rows, which confir
13 -----
14 --This means there were no foreign key dep
15 --Therefore, the brand could be safely rem
16 select brand_name, brand_id
17 from production.brands
18 where brand_name = 'Santa Cruz'
19
```

100 % No issues found

Results Messages

	brand_id	product_name
183	1	Electra Townie Commute 8D Ladies' ...
184	1	Electra Townie Original 1 Ladies' - 20...
185	1	Electra Townie Original 21D EQ Ladi...
186	1	Electra Townie Original 21D Ladies' -...
187	9	Santa Cruz Bronson

100 % No issues found

Results Messages

	brand_name	brand_id
1	Santa Cruz	9

```
SQLQuery1.sql...ujitsu (70)) *
9
10 SELECT brand_id, product_name
11 FROM production.products
12 --The query returned no rows, which confirmed that no products were linked to brand_id = 9.
13 -----
14
15 select brand_name, brand_id
16 from production.brands
17 where brand_name = 'Santa Cruz'
18 -----
19 --This means there were no foreign key dependencies preventing the deletion.
20 --Therefore, the brand could be safely removed using:
21
22 DELETE FROM production.brands
23 WHERE brand_name = 'Santa Cruz';
24
```

100 % No issues found Ln: 22, Ch: 30 TABS C

Results Messages

brand_name	brand_id
------------	----------

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7\ujitsu... BikeStores 00:00:00 Row: 1, Col: 1

--This means there were no foreign key dependencies preventing the deletion.

--Therefore, the brand could be safely removed using:

-- Q22: Delete all order_items that have quantity = 0.

SQLQuery1.sql...ujitsu (70))*

```
1
2 SELECT *
3 FROM sales.order_items
4 WHERE quantity = 0;
5
6
7 -- No action required
8 -- There are no records with quantity = 0 in sales.order_items
```

100 % No issues found

Results Messages

order_id	item_id	product_id	quantity	list_price	discount
----------	---------	------------	----------	------------	----------

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fujits...

-- Q23: Delete all customers who have never placed an order (use subquery with NOT EXISTS).

```
SQLQuery1.sql...ujitsu (70))*
1
2
3 SELECT *
4 FROM sales.customers c
5 WHERE NOT EXISTS (
6     SELECT 1
7     FROM sales.orders o
8     WHERE o.customer_id = c.customer_id
9 );
10
11 --Conclusion:
12 --0 rows returned
13 --No Action Required
14 -- It means:
15
16 --Every customer has placed at least one order
17 --There are no "orphan customers"
18 --So nothing qualifies for deletion
```

100 % No issues found

customer_id	first_name	last_name	phone	email	street	city	state	zip_code	loyalty_program_id
-------------	------------	-----------	-------	-------	--------	------	-------	----------	--------------------

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7A

ANS : No customers were found without orders, so no deletion is necessary.

-- Q24: Delete all products that have list_price > 10000 and model_year < 2020.

```
SQLQuery1.sql...ujitsu (70))*
1
2
3  --safe Check before deletion:
4
5  SELECT *
6  FROM production.products
7  WHERE list_price > 10000
8         AND model_year < 2020;
9
10 -- Ans :
11
12 --SELECT = verify impact
13 --DELETE = execute only if needed
14 --0 rows = safe (nothing will be deleted)
15 -- No rows returned
16 -- No action required
17 -- There are no products with list_price > 10000 AND model_year < 2020
18 --Since the safe check returned no rows, no deletion is required for Q24.
```

100 % No issues found Ln: 18, Ch:

product_id	product_name	brand_id	category_id	model_year	list_price
------------	--------------	----------	-------------	------------	------------

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7\fujitsu... BikeStores 00:00:00

-- Q25: Delete the loyalty_programs table you created in Q6 (clean up).

SQLQuery1.sql...ujitsu (70))*

```
1
2
3  --Safe practice :
4  --check table exists
5  SELECT *
6  FROM sales.loyalty_programs;
7  -- Yes, It exists
8
9  --Delete Table [sales.loyalty_programs]
10 DROP TABLE sales.loyalty_programs;
```

100 %

✓ No issues found

Results Messages

program_id	program_name	discount_rate	start_date	end_date
------------	--------------	---------------	------------	----------

✓ Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM)

sales.loyalty_programs

- Columns
 - program_id (PK, int, not null)
 - program_name (varchar(100), not null)
 - discount_rate (decimal(3,2), not null)
 - start_date (date, not null)
 - end_date (date, null)
- Keys
- Constraints

```
SQLQuery1.sql...ujitsu (70))*
2
3 --Safe practice :
4 --check table exists
5 SELECT *
6 FROM sales.loyalty_programs;
7 -- Yes, It exists
8
9 --Delete Table [sales.loyalty_programs]
10 DROP TABLE sales.loyalty_programs;
11
12 --sales.loyalty_programs table kisi aur table ke saath
13 -- linked (referenced) hai via FOREIGN KEY
14 --Simple explanation:
15
16 --loyalty_programs => Parent table
17 --Linked another table (e.g. customers) = Child table
18 -- Child table referenced parentTable
19 -- so parent table can not delete
```

100 % No issues found

Messages

Msg 3726, Level 16, State 1, Line 10
Could not drop object 'sales.loyalty_programs' because it is referenced by a FOREIGN KEY constraint.

Completion time: 2026-05-26T10:47:36.8646064+05:00

100 % No issues found Ln: 5, Ch:

Query completed with errors. DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\ujitsu... | BikeStores

```
-- =====
-- SECTION F: COMBINED & CHALLENGE QUESTIONS
-- =====
```

- ```
-- Q26: Write a single transaction that:
--
-- 1. Creates a new store called 'Downtown LA'
--
-- 2. Adds 3 new staff members to that store
--
-- 3. Inserts 100 units of product_id = 1 into stocks for that store
--
-- (ROLLBACK if any step fails)
```

```
1 BEGIN TRY
2 BEGIN TRANSACTION;
3
4 DECLARE @store_id INT;
5
6 -- 1. Store insert kiya
7 INSERT INTO sales.stores (store_name)
8 VALUES ('Downtown LA');
9
10 -- 2. Sahi ID pakdi
11 SET @store_id = SCOPE_IDENTITY();
12
13 -- 3. Staffs mein inserts (Comments ko alag line mein kar diya hy taaki code block
14 INSERT INTO sales.staffs (first_name, last_name, email, store_id)
15 VALUES
16 ('John', 'Doe', 'johndoe.final@gmail.com', @store_id),
17 ('Jane', 'Smith', 'janesmith.final@gmail.com', @store_id),
18 ('Sara', 'Brown', 'sarabrown.final@gmail.com', @store_id);
19
```

100 %

No issues found

Ln: 26, Ch:

Results

Messages

|   | Status   | staff_id | first_name | email                     | store_id |
|---|----------|----------|------------|---------------------------|----------|
| 1 | Success! | 22       | John       | johndoe.final@gmail.com   | 13       |
| 2 | Success! | 23       | Jane       | janesmith.final@gmail.com | 13       |
| 3 | Success! | 24       | Sara       | sarabrown.final@gmail.com | 13       |

SQLQuery8.sql...ujitsu (60))\*
SQLQuery1.sql...fujitsu (70))\*

```

1
2
3
4
5 select * from sales.stores;
6
7 select * from sales.staffs;

```

100 %
No issues found
Ln: 7, Ch: 1

Results

Messages

|    | staff_id | first_name | last_name | email                         | phone          | active | store_id | manager_id |
|----|----------|------------|-----------|-------------------------------|----------------|--------|----------|------------|
| 3  | 3        | Genna      | Serrano   | genna.serrano@bikes.shop      | (831) 555-5556 | 1      | 1        | 5          |
| 4  | 4        | Virgie     | Wiggins   | virgie.wiggins@bikes.shop     | (831) 555-5557 | 1      | 1        | 5          |
| 5  | 5        | Jannette   | David     | jannette.david@bikes.shop     | (516) 379-4444 | 1      | 2        | 1          |
| 6  | 6        | Marcelene  | Boyer     | marcelene.boyer@bikes.shop    | (516) 379-4445 | 1      | 2        | 5          |
| 7  | 7        | Venita     | Daniel    | venita.daniel@bikes.shop      | (516) 379-4446 | 1      | 2        | 5          |
| 8  | 8        | Kali       | Vargas    | kali.vargas@bikes.shop        | (972) 530-5555 | 1      | 3        | 1          |
| 9  | 9        | Layla      | Terrell   | layla.terrell@bikes.shop      | (972) 530-5556 | 1      | 3        | 7          |
| 10 | 10       | Bernardine | Houston   | bernardine.houston@bikes.shop | (972) 530-5557 | 1      | 3        | 7          |
| 11 | 22       | John       | Doe       | john.doe.final@gmail.com      | NULL           | 1      | 13       | NULL       |
| 12 | 23       | Jane       | Smith     | jane.smith.final@gmail.com    | NULL           | 1      | 13       | NULL       |
| 13 | 24       | Sara       | Brown     | sara.brown.final@gmail.com    | NULL           | 1      | 13       | NULL       |

Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM)

DESKTOP-26P7AE7\ujitsu...

BikeStor

-- Q27: Change the schema of sales.order\_items: add a new column 'tax\_amount' DECIMAL(8,2) DEFAULT 0.00,

-- then update it to be (list\_price \* quantity \* discount \* 0.08) for all existing rows.

SQLQuery8.sql...fujitsu (60))\*

SQLQuery1.sql...ujitsu (70))\*

```
1 BEGIN TRY
2 BEGIN TRANSACTION;
3
4 -- Step 1: sales.order_items table mein naya column add karein
5 IF NOT EXISTS (
6 SELECT * FROM INFORMATION_SCHEMA.COLUMNS
7 WHERE TABLE_SCHEMA = 'sales' AND TABLE_NAME = 'order_items' A
8)
9 BEGIN
10 ALTER TABLE sales.order_items
11 ADD tax_amount DECIMAL(8,2) NOT NULL DEFAULT 0.00;
12 END
13
14 -- Step 2: Dynamic SQL ke zariye update chalana taaki 'Invalid Co
15 EXEC sp_executesql N'
16 UPDATE sales.order_items
17 SET tax_amount = CAST((list_price * quantity * discount * 0.0
18 ' ;
19
```

100 %

No issues found

Messages

(3487 rows affected)  
Schema changed and tax\_amount updated successfully!  
  
Completion time: 2026-05-26T12:08:54.4718537+05:00

100 %

No issues found

Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fujit

SQLQuery8.sql...ujitsu (60))\*
SQLQuery1.sql...fujitsu (70))\*

```

1
2
3
4
5 select * from sales.order_items;

```

100 %
No issues found

Results

Messages

|    | order_id | item_id | product_id | quantity | list_price | discount | tax_amount |
|----|----------|---------|------------|----------|------------|----------|------------|
| 1  | 1        | 1       | 20         | 1        | 599.99     | 0.20     | 9.60       |
| 2  | 1        | 3       | 10         | 2        | 1549.00    | 0.05     | 12.39      |
| 3  | 1        | 4       | 16         | 2        | 599.99     | 0.05     | 4.80       |
| 4  | 2        | 1       | 20         | 1        | 599.99     | 0.07     | 3.36       |
| 5  | 2        | 2       | 16         | 2        | 599.99     | 0.05     | 4.80       |
| 6  | 3        | 1       | 3          | 1        | 999.99     | 0.05     | 4.00       |
| 7  | 3        | 2       | 20         | 1        | 599.99     | 0.05     | 2.40       |
| 8  | 4        | 1       | 2          | 2        | 749.99     | 0.10     | 12.00      |
| 9  | 5        | 1       | 10         | 2        | 1549.00    | 0.05     | 12.39      |
| 10 | 5        | 2       | 17         | 1        | 429.00     | 0.07     | 2.40       |
| 11 | 5        | 3       | 26         | 1        | 599.99     | 0.07     | 3.36       |

Query executed successfully.
DESKTOP-26P7AE7 (17.0 RTM)
DESKT

-- Q28: Identify and delete duplicate email addresses in sales.customers (keeping the smallest customer\_id).

SQLQuery8.sql...fujitsu (60))\* SQLQuery1.sql...ujitsu (70))\*

2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20

```
BEGIN TRY
BEGIN TRANSACTION;

-- CTE banayein jo har email ko partition kar ke unhein numbering dega
WITH CTE_Duplicates AS (
 SELECT
 customer_id,
 email,
 -- partition by email se har email ka alag group banega
 -- order by customer_id se sabse choti ID ko number 1 milega
 ROW_NUMBER() OVER (PARTITION BY email ORDER BY customer_id ASC) AS row_n
 FROM
 sales.customers
 WHERE
 email IS NOT NULL -- Khali emails ko ignore karne ke liye
)
-- Jo number 1 se bade hain (yaani duplicates), unhein delete kar dein
DELETE FROM CTE_Duplicates
```

100 % No issues found Ln: 29, 1

Messages

(0 rows affected)

Duplicate emails deleted successfully, keeping the smallest customer\_id!

Completion time: 2026-05-26T12:11:59.1282326+05:00

100 % No issues found Ln: 6, Ch: 1 TABS

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7\fujits... BikeStores 00:00:01

SQLQuery8.sql...ujitsu (60))\* SQLQuery1.sql...fujitsu (70))\*

1  
2  
3  
4  
5 select \* from [sales].[customers];

100 % No issues found Ln: 5, Ch: 35 TABS CRLF UTF

Results Messages

|    | customer_id | first_name | last_name | phone          | email                       | street                  | city            | state | zip_code | loyalty_program_id |
|----|-------------|------------|-----------|----------------|-----------------------------|-------------------------|-----------------|-------|----------|--------------------|
| 1  | 1           | Debra      | Burks     | NULL           | debra.burks@yahoo.com       | 9273 Thome Ave.         | Orchard Park    | NY    | 14127    | NULL               |
| 2  | 2           | Kasha      | Todd      | NULL           | kasha.todd@yahoo.com        | 910 Vine Street         | Campbell        | CA    | 95008    | NULL               |
| 3  | 3           | Tameka     | Fisher    | NULL           | tameka.fisher@aol.com       | 769C Honey Creek St.    | Redondo Beach   | CA    | 90278    | NULL               |
| 4  | 4           | Daryl      | Spence    | NULL           | daryl.spence@aol.com        | 988 Pearl Lane          | Uniondale       | NY    | 11553    | NULL               |
| 5  | 5           | Charolette | Rice      | (916) 381-6003 | charolette.rice@msn.com     | 107 River Dr.           | Sacramento      | CA    | 95820    | NULL               |
| 6  | 6           | Lyndsey    | Bean      | NULL           | lyndsey.bean@hotmail.com    | 769 West Road           | Fairport        | NY    | 14450    | NULL               |
| 7  | 7           | Latasha    | Hays      | (716) 986-3359 | latasha.hays@hotmail.com    | 7014 Manor Station Rd.  | Buffalo         | NY    | 14215    | NULL               |
| 8  | 8           | Jacqueline | Duncan    | NULL           | jacqueline.duncan@yahoo.com | 15 Brown St.            | Jackson Heights | NY    | 11372    | NULL               |
| 9  | 9           | Genoveva   | Baldwin   | NULL           | genoveva.baldwin@msn.com    | 8550 Spruce Drive       | Port Washington | NY    | 11050    | NULL               |
| 10 | 10          | Pamelia    | Newman    | (555) 123-4567 | pamelia.newman@gmail.com    | 476 Chestnut Ave.       | Monroe          | NY    | 10950    | NULL               |
| 11 | 11          | Deshawn    | Mendoza   | NULL           | deshawn.mendoza@yahoo.com   | 8790 Cobblestone Street | Monsey          | NY    | 10952    | NULL               |

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7/fujitsu... BikeStores 00:00:00 Row: 1, Col: 1 1,445 rows

-- Q29: Archive all orders from year 2020 or older:

-- Insert them into a new table sales.orders\_archive, then delete from sales.orders.

SQLQuery9.sql... Executing...\*

SQLQuery1.sql...ujitsu (70))\*

```

1 BEGIN TRY
2 BEGIN TRANSACTION;
3
4 -- Step 1: Agar sales.orders_archive table pehle se nahi bana hua, to use create karein
5 IF NOT EXISTS (SELECT * FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA = 'sales' AND
6 BEGIN
7 SELECT * INTO sales.orders_archive
8 FROM sales.orders
9 WHERE 1 = 0;
10 END
11
12 -- Step 2: IDENTITY_INSERT ko ON karein taaki hum purani IDs archive table mein bhej sa
13 SET IDENTITY_INSERT sales.orders_archive ON;
14
15 -- Step 3: Saare columns ke naam likh kar insert karein
16 -- (Note: Agar aapke table mein iske alawa columns hain to unhe bhi is list mein shamil
17 INSERT INTO sales.orders_archive (
18 order_id, customer_id, order_status, order_date,
19 required_date, shipped_date, store_id, staff_id

```

100 % No issues found Ln: 16, Ch: 46 SP

Messages

(1615 rows affected)

(3487 rows affected)

(1615 rows affected)

All orders from 2020 or older have been archived and cleaned successfully!

100 % No issues found Ln: 4, Ch: 21 TABS MIXED

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM) DESKTOP-26P7AE7fujitsu... BikeStores 00:00:00 Row: 1,

100 % 1 0

Results Messages

|   | Archived_Orders_Count |
|---|-----------------------|
| 1 | 1615                  |

-- Q30: Add a CHECK constraint to production.products ensuring list\_price >= 0 AND model\_year BETWEEN 1900 AND YEAR(GETDATE())+1.

SQLQuery2.sql...fujitsu (57))\*

SQLQuery1.sql...ujitsu (55))\*

```
1
2 BEGIN TRY
3 BEGIN TRANSACTION;
4
5 -- production.products table par CHECK constraint lagayein
6 ALTER TABLE production.products
7 ADD CONSTRAINT CHK_Products_Price_Year
8 CHECK (list_price >= 0 AND model_year BETWEEN 1900 AND 2027)
9
10 COMMIT TRANSACTION;
11 PRINT 'CHECK constraint successfully added!';
12 END TRY
13 BEGIN CATCH
14 IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
15 PRINT 'Error Occurred: ' + ERROR_MESSAGE();
16 END CATCH;
```

00 % No issues found

Messages

CHECK constraint successfully added!

Completion time: 2026-05-26T14:05:15.8131209+05:00

00 % No issues found

Ln: 1,

Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7A

SQLQuery2.sql...ujitsu (57))\* SQLQuery1.sql...fujitsu (55))\*

1  
2  
3  
4  
5  
6  
7  
8  
9  
10

SELECT  
    name AS Constraint\_Name,  
    definition AS Constraint\_Definition,  
    is\_disabled AS Is\_Disabled  
FROM  
    sys.check\_constraints  
WHERE  
    name = 'CHK\_Products\_Price\_Year';

100 % No issues found

Results Messages

|   | Constraint_Name         | Constraint_Definition                              | Is_Disabled |
|---|-------------------------|----------------------------------------------------|-------------|
| 1 | CHK_Products_Price_Year | ([list_price]>=(0) AND ([model_year]>=(1900) AN... | 0           |

Query executed successfully. DESKTOP-26P7AE7 (17.0 RTM)

-- =====

-- END OF HOMEWORK QUESTIONS

-- =====